



Fangxiao Formula alleviates airway inflammation and remodeling in rats with asthma via suppression of transforming growth factor- β /Smad3 signaling pathway

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ABSTRACT

Asthma is a common obstructive airway disease characterized by inflammation and remodeling with a progressive decline in lung function. Fangxiao Formula (FXF) is an herbal medicine that has achieved significant clinical benefits toward asthma patients, but the relevant mechanism has not yet been clarified. The aim of this study was to determine the inhibitory effects of FXF on airway inflammation and remodeling, and investigate the activities of TGF- β /Smads signaling pathway in the rat asthma model. Rats were sensitized by ovalbumin (OVA) for six weeks to establish the asthma experimental model. OVA-challenged animals were randomly divided into 5 groups and received different concentrations of FXF or dexamethasone. The animals in blank control group received saline only. Lung tissues were collected and analyzed for determining the inflammatory cells infiltration, HE and PAS staining, airway wall thickness and collagen deposition. The productions of inflammatory cytokine productions were analyzed by ELISA in the bronchoalveolar lavage (BAL) fluid. Immunohistochemical analysis was performed to measure the expression of α -SMA and PCNA in lung tissue after the treatment of FXF. The levels of TGF- β were assessed by both immunohistochemistry and western blotting, and the expression of p-Smad2/3 proteins were determined by western blotting analysis. Our results indicated that FXF attenuated the infiltration of inflammatory cells, decreased the production of Th2 cytokines and simultaneously increased the levels of Th1 cytokine in the asthma rat model. In addition, FXF reduced allergen-induced increased airway wall thickness, goblet cell hyperplasia and collagen deposition. Furthermore, the expression levels of TGF- β and p-Smad3 were obviously reduced after the treatment of FXF. These results indicate that FXF alleviates airway inflammation and remodeling by restoring the balance of Th1/Th2 cytokines and the TGF- β /Smad-3 pathway, therefor providing potential therapeutic approach for asthmatic patients.

1. Introduction

Asthma is a chronic inflammatory disorder of the airway involving multiple cells and cellular components, which is characterized by airway hyperresponsiveness (AHR), persistent airway inflammation and airway remodeling [1]. Since the significantly increasing morbidity and mortality, asthma has become a severe global public health problem worldwide. In China, the total incidence of childhood asthma in cities was 3.02%, which are increasing steadily year by year [2]. Because of the complexity of asthma etiology and pathogenesis, the clinical treatments have also met great challenges. Long-term anti-inflammatory treatments with glucocorticoids are the standard

therapeutic approach for asthma [3]. However, the side effects of current treatment were common and may be problematic, involving the musculoskeletal, cardiovascular, neuropsychiatric, gastrointestinal and immunologic in nature. Thus, there is an urgent need to investigate and develop novel strategies for asthma with minimal side effects.

Herbal medicines with multiple components and ubiquitous pharmacological activities have been widely used in the treatment for various complex diseases [4,5]. Furthermore, treating asthma with herbs could achieve ideal therapeutic effects with minor side effects [6]. The traditional herbal medicine Fangxiao Formula (FXF) is composed of eleven medicinal herbs including *Radix adenophorae*, *Radix glehniae*, *Radix codonopsis*, *Radix ophiopogonis*, *Cortex lyciiradicis*, *Radices*

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paeoniae alba, *Concha haliotidis*, *Poria cocos*, *Atractylodes macrocephala*, *Pinellia ternata* and *Pericarpium citri reticulatae*. FFX has been used for the treatment of asthma in children and shown clinical benefits. In addition, extracts of *Radix adenophorae*, the main active herbs of FFX, suppressed the development on inflammation and decreased airway damage in the OVA-induced model of asthma [7]. Our previous studies also demonstrated that FFX has the potential to alleviate asthma symptoms in OVA-induced asthma rats [8]. Despite the potential therapeutic benefit of FFX in the treatment of asthma, the underlying mechanism of FFX on airway inflammation and remodeling remains unknown.

The chronic airway inflammation and airway remodeling are the distinctive characteristics of asthma [9]. In asthmatic patients, the increased inflammatory cells of eosinophils and T lymphocytes infiltrate peribronchial lung tissue, leading to the structural changes in the airway and collectively termed airway remodeling. Airway remodeling contributed to the persistent airway hyper-responsiveness and induced an irreversible airway obstruction, leading to the obstinate asthma with pulmonary function depression [10]. Recent studies suggest that targeting airway inflammation and remodeling may be therapeutic approaches for asthma, providing potential agents for the treatment of asthma [11,12]. It has been shown that inhibition of airway inflammation by Brg1 gene knock-out could lead to the amelioration on house dust mite induced asthma [13]. Paraoxonase-1 relived airway inflammation and remodeling in asthmatic mice via inhibiting the secretion of LPS-induced macrophage inflammatory cytokines [14]. The extracts from *Astragalus membranaceus* inhibited asthma airway remodeling by down-regulating the activity of the TGF- β /Smad pathway, providing novel therapeutic strategy for asthma patients [11]. Furthermore, sinomenine could also improve airway inflammation and remodeling through targeting TGF- β and CTGF pathways [15]. Thus, search for bioactive natural products targeting airway inflammation and remodeling would contribute to the treatment of asthma.

In this study, we find that the traditional herbal medicine Fangxiao Formula (FFX) could effectively ameliorate both airway inflammation and structural changes in OVA sensitized asthma rat model. These therapeutic effects were mediated by inhibiting the activity TGF- β /Smad signaling pathway, in turn suppressing the secretion of Th2 cells and infiltration of inflammatory cell into lung tissue, and reducing the airway thickness, goblet cell hyperplasia and collagen deposition. Our results demonstrate the potential benefits of FFX against toward airway inflammation and remodeling associated with bronchial asthma.

2. Materials and methods

2.1. Preparation of FFX and the fingerprint

Fangxiao Formula (FFX) was provided by the first affiliated hospital of Zhejiang Chinese medical university. The 11 Chinese medicinal ingredients were mixed in a certain proportion (Table 1), soaked with

Table 1
The composition and ratio of eleven herbs used in the preparation of Fangxiao Formula.

Pharmaceutical name	Chinese name	Part used	Ratio
Radix Adenophorae	Nanshashen	Root	6
Radix Glehniae	Beishashen	Root	6
Radix Codonopsis	Dangshen	Root	6
Radix Ophiopogonis	Maidong	Root	9
Cortex Lycii Radicis	Digupi	Root bark	3
Radices Paeoniae Alba	Baishao	Root	6
Concha Haliotidis	Shijueying	Abalone shell	10
Poria Cocos	Fuling	Sclerotium	10
Atractylodes Macrocephala	Baizhu	Root and Rhizome	6
Pinellia Ternata	Banxia	Tuber	6
Pericarpium Citri Reticulatae	Chenpi	Pericarpium	5

twenty folds volume of water for 30 min, and boiled for 1 h. After boiling twice, the FFX decoction was pooled together and then evaporated to dryness with vacuum freeze drying (obtain final 12.80% yields). The powder was dissolved with double distilled water at different concentrations (5 mg/mL, 10 mg/mL and 20 mg/mL, respectively).

The fingerprinting of FFX was carried out by high performance liquid chromatography (HPLC) to control the stability of FFX decoction (Fig. 1). The 10 batches of FFX decoction were collected respectively and analyzed after the centrifugation and filtration. Chromatographic separation was performed using a Waters Breeze system consisting with an autosampler 2707 and binary HPLC pump 1525, and equipped with an XBridge C18 column (Waters, No. 186003117) with Waters 2489 UV/Visible detector and Empower software. The acetonitrile-water mobile phase system with the flow rate at 0.5 mL/min and the column temperature of 25 °C was performed on the reversed phase liquid chromatography. The designed gradient elution program was as follows: 0% of acetonitrile (0–10 min); 0–20% of acetonitrile (10–20 min); 20–50% of acetonitrile (20–35 min); 50–100% of acetonitrile (35–50 min) and 100–0% of acetonitrile (50–60 min). The putative compounds of FFX decoction were determined by comparing their retention time with those of standard marker.

2.2. Rats

Male SD (Sprague–Dawley) rats (specific pathogen free grade, four weeks old) were purchased from the Laboratory Animal Center of Zhejiang Chinese Medical University. All experimental procedures were conformed to the protocol approved by the Animal Ethics Committee of Zhejiang Chinese Medical University. Sixty animals were divided randomly into six experimental groups of 10 animals each: normal control group (received equivalent volumes of physiological saline, 10 mL/kg), model group (received physiological saline, 10 mL/kg), positive group (received Dexamethasone, 0.5 mg/kg), FFX-low group (FFX-L, 50 mg/kg), FFX-medium group (FFX-M, 100 mg/kg) and FFX-high group (FFX-H, 200 mg/kg). As shown in Fig. 2, except for the normal control group, rats at other five groups were actively sensitized on protocol days 0 and 14 by two intraperitoneal injections with 10 mg OVA (egg albumin grade II; Sigma) plus 100 mg of aluminum hydroxide gel, respectively. From day 15, the rats were placed into a closed chamber (35 × 25 × 15 cm³) for exposure to aerosol of 2% OVA for three consecutive weeks (30 min/time and 1 time/2 day). Two hours before challenging with OVA atomization, rats were treated with different concentrations of FFX (FFX-L, FFX-M and FFX-H groups) by intragastric administration once a day. Rats treated with 0.5 mg/kg Dexamethasone by intraperitoneal injections were used as the drug control.

2.3. Tissue preparation

Within 24 h after the last OVA challenge, each rat was intraperitoneally injected with 10% chloral hydrate at a dose of 4 mg/kg. In the rat open thoracic cavity, the upper segments of the trachea were cut and cannulated. The left lungs were ligated at the left main bronchus, and the right lungs were lavaged through the bronchus. Then 5 mL of phosphate buffer solution was slowly infused and extracted three times. The collected bronchoalveolar lavage (BAL) fluid was centrifuged at 300 g for 10 min, and the supernatants were preserved at –80 °C until use. The BALF sediment was re-suspended in PBS for cytospin so that leukocyte counts could be obtained. After collection of BALF, lung tissues were cut from the right lungs and fixed in 4% paraformaldehyde, followed by paraffin embedding. Additional lung tissue samples were stored in liquid nitrogen for protein extraction.

2.4. Total and differential BAL fluid leukocyte counts

Total and differential cell counts were performed using sedimentary

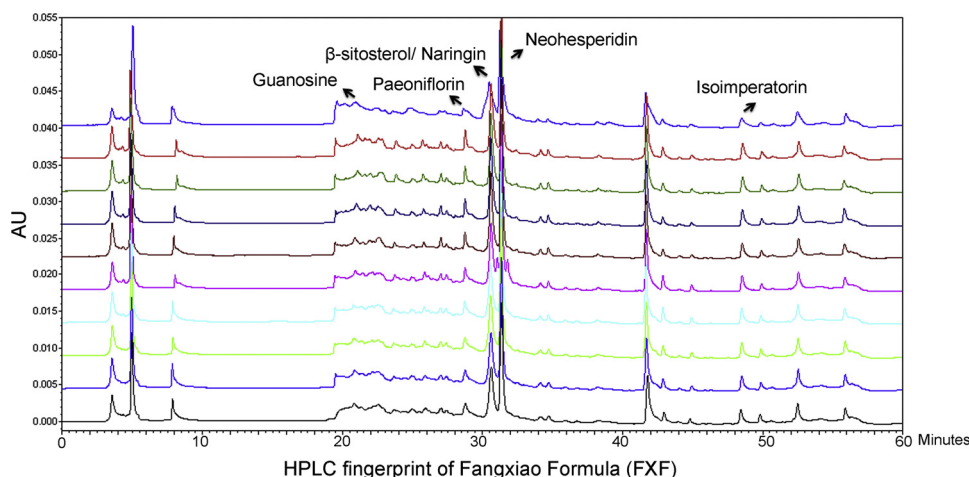


Fig. 1. HPLC fingerprint of Fangxiao Formula (FXF). The HPLC analysis from ten different batches of FXF was performed to guarantee the FXF quality and batch parallelism.

smears within 60 min after BAL fluid obtained. The cells suspension were fixed on clean glass slides and stained with Wright-Giemsa solution. The eosinophils were differentiated and counted according to their morphological criteria. Usually 200 white blood cells are counted and the percentage of eosinophils is calculated.

2.5. Hematoxylin-eosin (H&E) and PAS staining

The Rat lung tissues were fixed in 4% paraformaldehyde 3 days, and then embedded with paraffin. Sections (5- μ m thick) were stained with H&E using standard histological techniques. After H&E stained, the pulmonary tissue sections were observed pathological and structure changes under the optical microscope. As described by Myou et al, the inflammation injury around bronchus was semi-quantitative analyzed and graded on the 5 scales: 0, normal; 1, few cells; 2, a ring of inflammatory cells 1 cell layer deep; 3, a ring of inflammatory cells 2–4 cells deep; 4, a ring of inflammatory cells > 4 cells deep [16]. The paraffin-embedded lung tissues were also sectioned (5- μ m thick) and conducted with periodic acid-Schiff (PAS) staining to identify goblet cells in the epithelium of airway. As described by Padrid et al, goblet cell hyperplasia was reflected by PAS score, which was counted the percentage of PAS-positive cells among all bronchial epithelial cells: 0, no goblet cells; 1, < 25%; 2, 25%–50%; 3, 51%–75%; 4, > 75% [17].

2.6. Immunohistochemistry

The paraffin-embedded Rat lung tissues were sectioned (5- μ m thick) and the α -SMA (Anti-alpha smooth muscle Actin Antibody, #BM3902, Boster Biological Technology Co., Ltd.), PCNA (Anti-PCNA Antibody, #BM0104, Boster Biological Technology Co., Ltd.), collagen I (Anti-Collagen I (COL1A1) Antibody, #BA0325, Boster Biological Technology Co., Ltd.), collagen III (Anti-Collagen III (COL3A1) Antibody, #SAB4500367, Sigma-Aldrich), collagen IV (Anti-Collagen IV (COL4A1)

Antibody, #SAB4500369, Sigma-Aldrich) and TGF- β 1 (Anti-active TGF- β 1 Antibody, #BA0290, Boster Biological Technology Co., Ltd.) protein levels were detected by immunohistochemistry and semi-quantified using Image Pro-Plus version 6.0 as we described previously [18]. The background staining with Rabbit IgG isotype antibodies of collagen type I, III, IV and TGF- β 1 was used as the negative control (Supplementary Fig. 1).

The thickness of the airway smooth muscle layer was scored according to the ratio of α -SMA positive area (μm^2) / the bronchiolar basement membrane (μm). The smooth muscle hyperplasia was quantified by counting the percentage of PCNA positive cells among the airway smooth muscle cells [12]. More than 5 complete bronchioles with medium and small size (200–400 μm) were quantified in each rat and 8 rats were analyzed in each group.

For collagen protein levels analysis, three complete bronchioles with medium and small size (150–400 μm) per section were measured under 200 $\times$ objective lens. The three collagen proteins were quantified according to the staining area just in the structural cells of the airways. The optical density values were quantified by Image Pro-Plus. Values from the chosen bronchioles image regions for each tissue slide were averaged.

2.7. Enzyme-linked immunosorbent assay (ELISA)

The levels of interleukin IL-4, IL-5, IL-9 and IL-13 were assessed in the supernatant of Bronchoalveolar lavage fluid using ELISA kits (Boster Biological Technology, China) according to the manufacturer's instructions.

2.8. Western blotting

Total proteins were extracted by Total Protein Extraction Kit (Applygen Technologies Inc., China) according to the manufacturer's

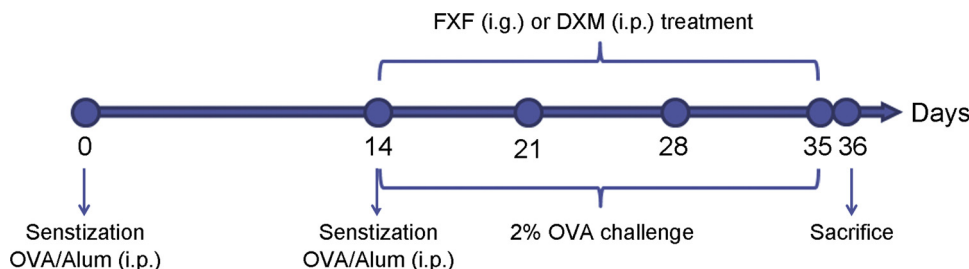


Fig. 2. Rat models of OVA-induced asthma and treatment with FXF and DXM.

instruction. Briefly, 50 mg frozen lung tissue was homogenated in 0.5 mL Lysis buffer and let stand for 10 min in 4 °C. Double volume Extraction buffer was added in to each sample and centrifugated at 10,000 g for 10 min. Then the intermediate layer was harvested and the protein content was measured by the BCA assay kit. About twenty-five micrograms total protein was prepared for western blotting as we described previously [19]. The antibodies of phospho-Smad2 (Ser465/467) (#3108) and phospho-Smad3 (Ser423/425) (#9520) were purchased from Cell Signaling Technology. The relative densities of the blots were semi-quantity analyzed by gray intensity using the Gel-Pro Analyzer. At least three times for each western blot experiment were performed and density data were collected from the average.

2.9. Statistical analysis

The data statistical analysis was performed by software of SPSS 17.0. One-way analysis of variance (ANOVA) was used to access the statistical significance between all groups and Tukey's to test significance between specific groups. $P < 0.05$ was considered as statistically significant differences between different groups.

3. Results

3.1. Inhibitory effects of FXF on airway inflammation in OVA-induced asthma model

The inflammatory cell infiltration was a typical feature of asthma. To obtain evidence for inflammatory cell infiltration into the tissue, the

rat lung sections were analyzed by HE staining after the last OVA challenge. As shown in Fig. 3A, compared with the control group, the number of inflammatory cells of the peribronchiolar and perivascular connective tissues in OVA-exposed rats significantly increased, suggesting obvious inflammatory cell infiltration into the lung tissue. In contrast, FXF treatment remarkably decreased the inflammatory cell infiltration at the concentration of 100 and 200 mg/kg administration (Fig. 3B). To further investigate that effect of FXF on inflammatory cells recruitment in lung tissues, we counted the number of inflammatory cells in bronchoalveolar lavage (BAL) fluid with microscopic images. The infiltrated cells were increased significantly in OVA-challenged rats compared with control group (Fig. 3C). The increased inflammatory cells were obviously decreased by administration of FXF. After the treatment of 100 mg/kg and 200 mg/kg FXF, the numbers of WBC infiltrating BALF decreased by 34.7% and 63.1%, respectively (Fig. 3C). Furthermore, FXF significantly inhibited the numbers of macrophages (MAC), eosinophils (EOS), lymphocytes (LYM) and neutrophils (NEU) infiltrating BALF in OVA challenged rats at a dose-dependent manner. These results suggested that FXF has an efficacy to suppress airway inflammation in the rat model with chronic asthma.

In order to determine the effect of FXF on cytokine production, the levels of four Th2 cytokines of IL-4, IL-5, IL-9 and IL-13 in the BALF were measured by ELISA after the OVA final challenge. Compared with rats in the normal control group, the OVA-challenged rats presented much higher levels of inflammatory cells in the BALF (Fig. 4). Significantly lower levels of IL-4, IL-5, IL-9 and IL-13 in the BALF were observed in the FXF treated rats compared with that of the OVA challenge group (Fig. 4). The expression level of IL-4 was 199.26 pg/mL in

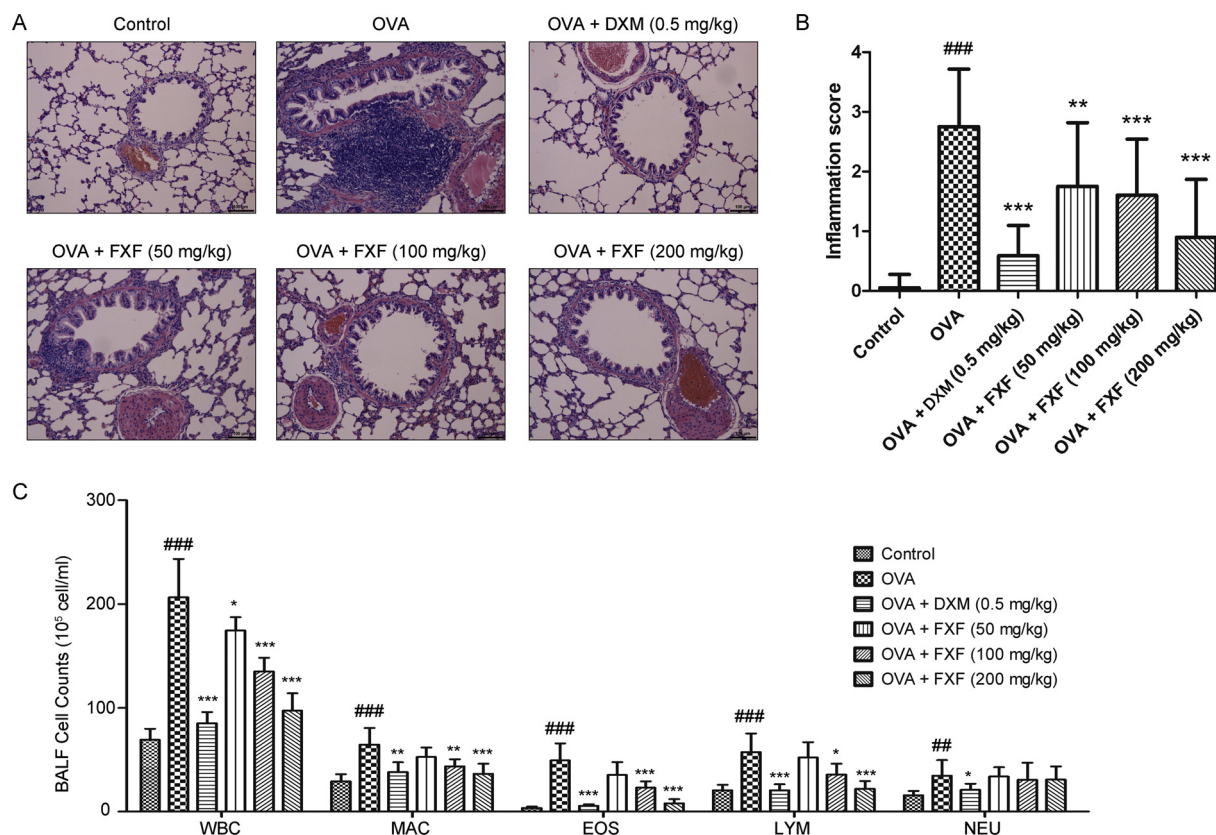


Fig. 3. FFX inhibited the recruitment of inflammatory cells in the lung tissues and bronchoalveolar (BAL) fluid. (A) Hematoxylin and eosin (HE) staining of lung tissues in OVA-challenged asthmatic rat. Lung tissue sections were obtained from the OVA induced rat and analyzed by HE staining. All photos were taken at x200 magnification. (B) The inflammation scores of lung tissue were calculated by the image analyzer. (C) Changes of total white cell as well as the other cell types (macrophages (MAC), eosinophils (EOS) and lymphocytes (LYM)) in the BALF after treatment of FFX. After the isolation of cytospin, the BALF cells were stained and analyzed with Wright-Giemsa staining reagent. Average data were represented as the mean $\pm$ standard from at least 3 separate experiments. ## $P < 0.01$ and ### $P < 0.001$ denote significant differences compared with the control group. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$ denote significant differences compared with the OVA-challenged group, respectively.

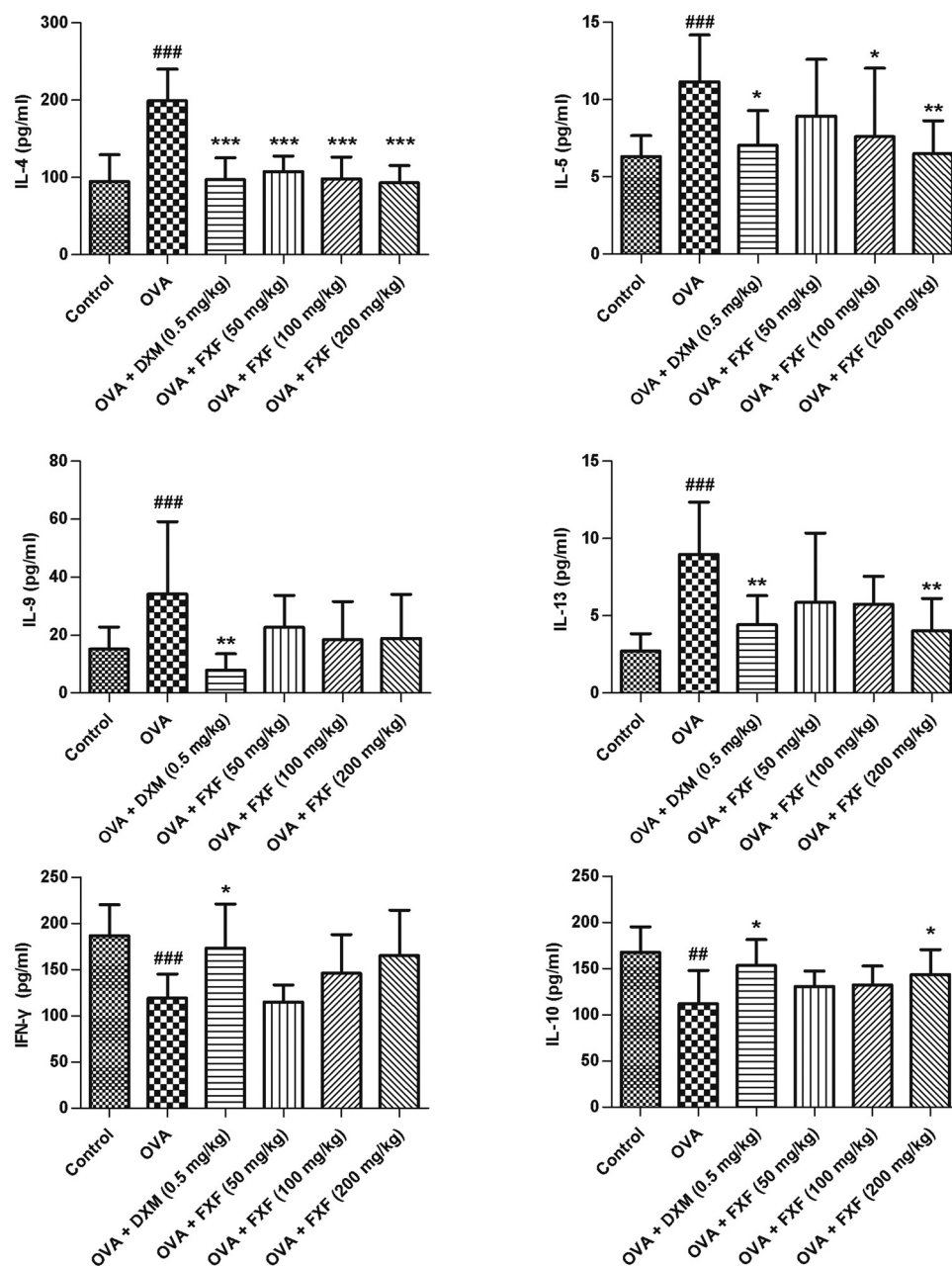


Fig. 4. Effect of FFX on the productions of Th1 and Th2 cytokines in the BALF. The expression levels of cytokines were determined using ELISA and indicated in (A) IL-4, (B) IL-5, (C) IL-9, (D) IL-13, (E) IFN- γ and (F) IL-10, respectively. Average data of cytokine productions were expressed as mean $\pm$ standard deviation ($n = 5$ per group). ## $P < 0.01$ and ### $P < 0.001$ denote significant differences compared with the control group. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$ denote significant differences compared with the OVA-challenged group, respectively.

the OVA challenged rats, while it reduced to 93.17 pg/mL after the treatment of 200 mg/kg FFX for 3 weeks (Fig. 4A). Furthermore, FFX also enhanced the amounts of Th1 cytokine IFN- γ as well as the anti-inflammatory cytokine IL-10 in the BALF, respectively (Fig. 4). These results demonstrated that FFX could inhibit the secretions of Th2 cytokines and increase the expression levels of Th1 cytokines, suggesting as a potential to restore the imbalanced ratio of Th1/Th2 in asthma.

3.2. Effects of FFX on the airway remodeling

Since the central role of airway remodeling in the process of asthma development, we determined the effects of FFX on airway smooth muscle (ASM) thickness and collagen deposition in lung tissue. Immunohistochemical staining was performed on the rat lung sections to evaluate the expression levels of α -SMA and PCNA (Fig. 5). The

expression levels of α -SMA significantly increased in the OVA-challenged rats, compared with that of the normal control group (Fig. 5A). After the treatment with FFX, the protein expression levels of α -SMA decreased obviously. The ASM thickness reduced by 44.1% and reached a similar size with that of the normal control group (Fig. 5A). Moreover, the expression levels of PCNA in the OVA-challenged rats were significantly much higher than that of the control group. After the treatment of DXM and high dose of FFX, the percentage of PCNA positive cells decreased to 14.8% and 21.6%, respectively, indicating an obvious improvement on the smooth muscle hyperplasia (Fig. 5B). These results demonstrated that FFX inhibited the airway remodeling in OVA sensitized rats through the mediation of α -SMA and PCNA protein levels.

The periodic acid-Schiff (PAS) staining was used to determine the number of goblet cells after FFX treatment. The OVA-challenged rats exhibited a remarkable increase on the number of goblet cell and mucus

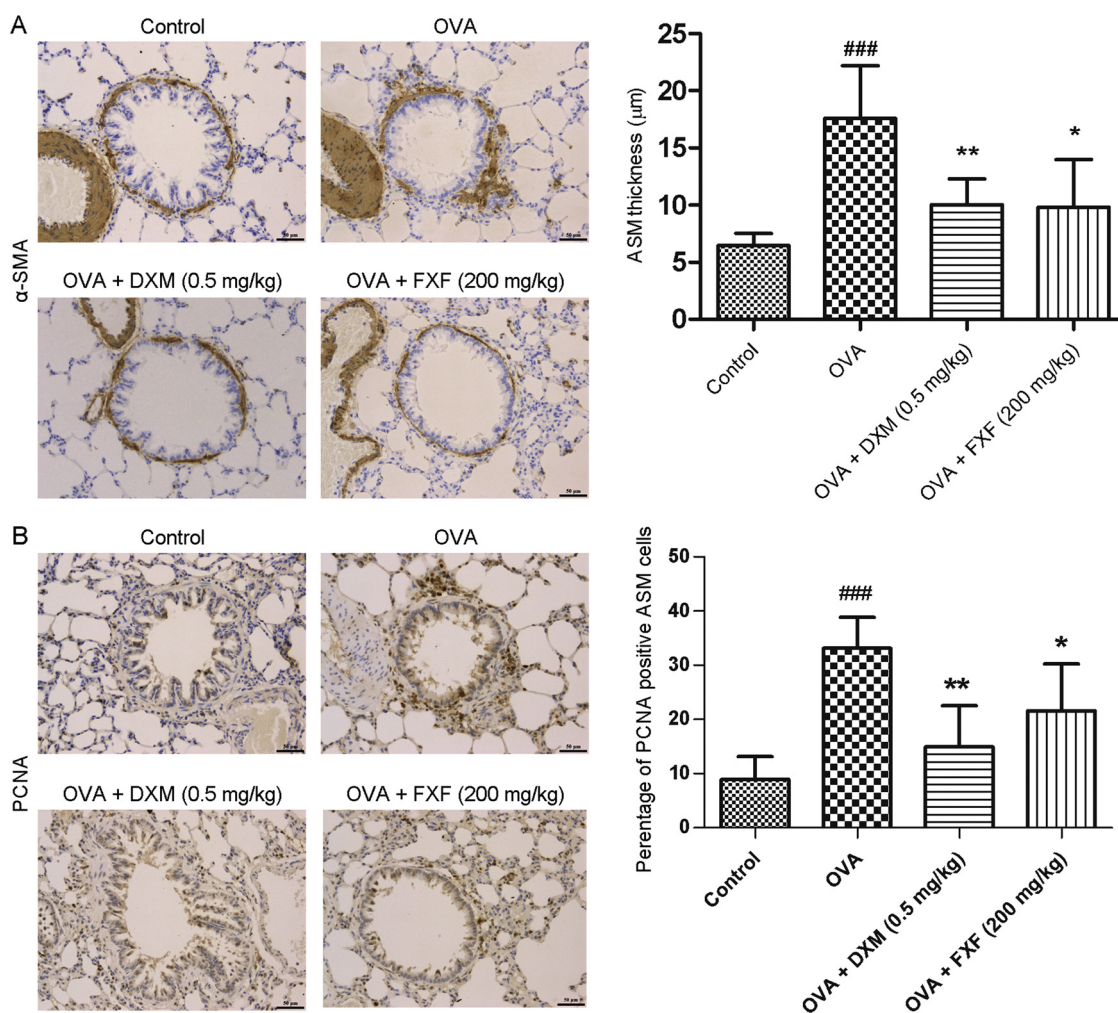


Fig. 5. FFX decreased the protein expressions of α -SMA and PCNA in the lung tissue from OVA-challenged rats. The expression levels of α -SMA (A) and PCNA (B) in each group were determined by immunohistochemical staining. The airway smooth muscle (ASM) thickness were measured according to the immunostained area of α -SMA per unit basement membrane length (μ m) in the bronchioles of each rat ($n = 8$ per group). ### $P < 0.001$ denotes significant differences compared with the control group. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$ denote significant differences compared with the OVA-challenged group, respectively.

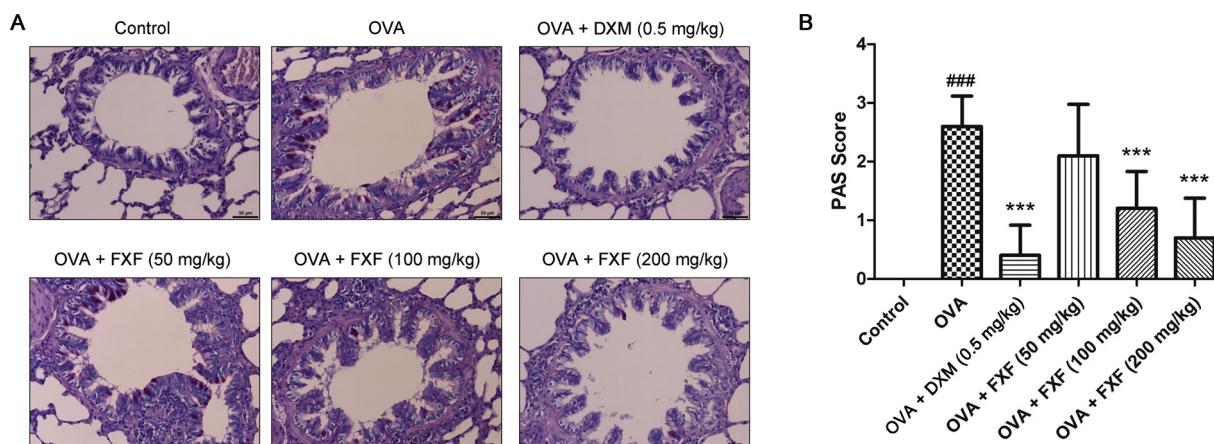


Fig. 6. FFX inhibited the airway goblet cell hyperplasia in lung tissues. (A) Periodic acid-Schiff (PAS) staining was performed on airway tissues from the OVA induced asthmatic rats. Lung tissue sections were obtained from the asthmatic rat and analyzed by PAS staining. All photos were taken at $\times 100$ magnification. (B) Goblet cell hyperplasia was determined by quantifying the percentage of goblet cells in the epithelium, which was scored according to the previous reports. The results are expressed in mean $\pm$ standard deviation from at 5 separate experiments. ### $P < 0.001$ denotes significant differences compared with the control group. *** $P < 0.001$ denotes significant differences compared with the OVA-challenged group.

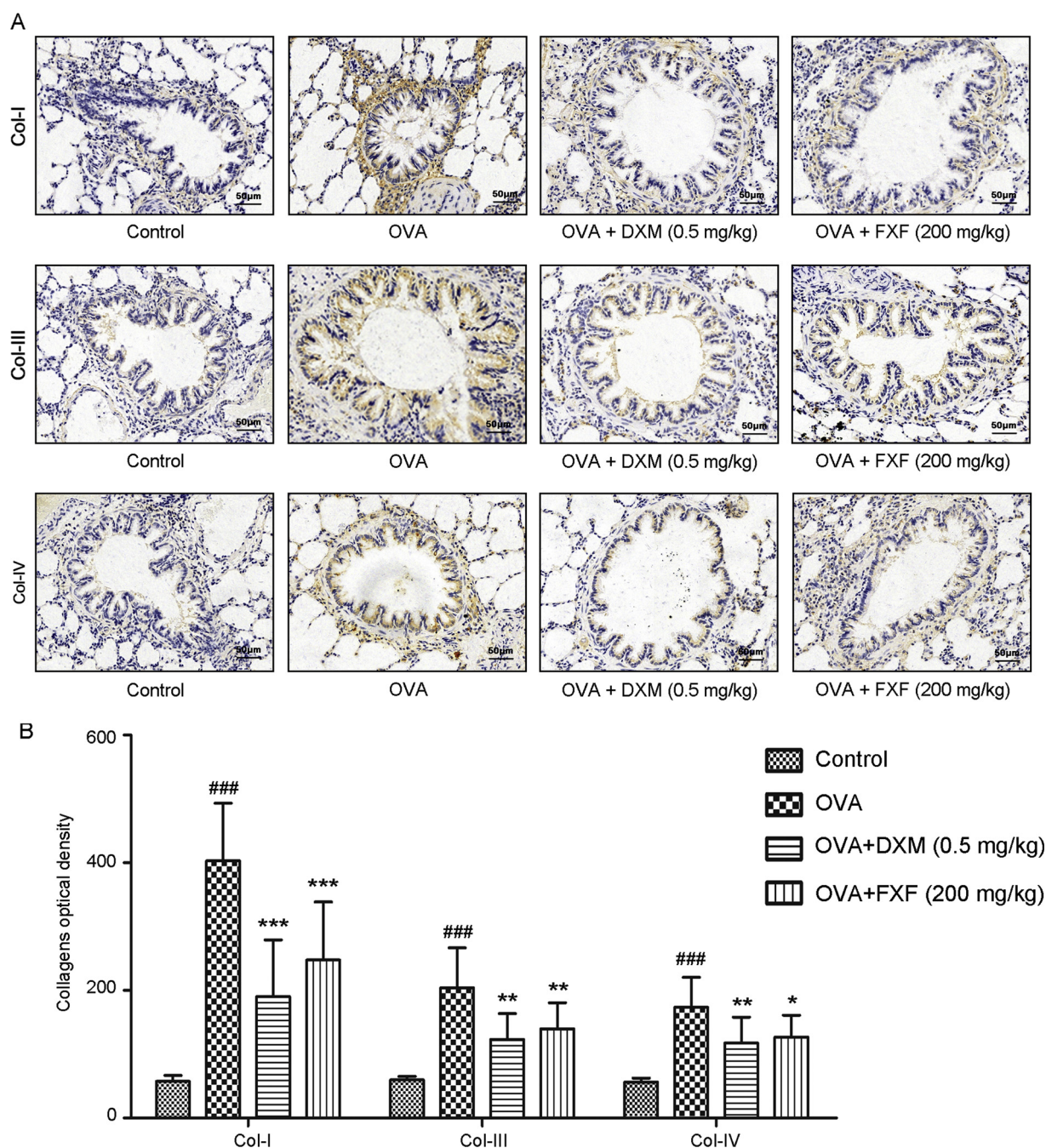


Fig. 7. FFX decreased the collagen deposition in lung tissue from OVA challenged asthmatic rat. (A) The expression levels of Col I, Col III and Col IV were analyzed by immunostaining from the lung tissue sections of asthmatic rats. (B) Semi-quantitative immunohistochemical staining analysis of Col I, Col III and Col IV in histological sections from rats after the treatment of FFX. The expression levels of collagen proteins were determined using the image analyzer. The quantification was limited to the structural cells of the airways. Average data were represented as the mean $\pm$ standard deviation ($n = 8$ per group). ### $P < 0.001$ denotes significant differences compared with the control group. *** $P < 0.001$ denotes significant differences compared with the OVA-challenged group.

secretion than that of the control group (Fig. 6). Compared with the OVA-challenged group, treatment with 100 mg/kg and 200 mg/kg FFX led to a significant decrease on the number of goblet cells with markedly lower PAS point score of 1.2 and 0.7, respectively (Fig. 6B). To further investigate the FFX on airway remodeling, we examined the deposition and intensity of collagen type I, III and IV with immunohistochemical staining analysis. As shown in Fig. 7, the collagen deposition significantly increased and extensively distributed in the airway wall surrounding in the OVA-challenged group (Fig. 7A). Compared with the OVA-challenged rats, rats treated with high dose of FFX or DXM exhibited much lower collagen deposition (Fig. 7B). After

the treatment of FFX on OVA-induced rats, the expressions of collagen type I, III and IV reduced by 38.6%, 31.5% and 26.8%, respectively, exhibiting similar results with that of DXM treatment group (Fig. 7B).

3.3. Influence of FFX on TGF- β /Smad signaling pathway

Since the central role of TGF- β signaling pathway toward airway inflammation and remodeling, the expression levels of TGF- β were determined using the rat lung sections from six groups. The OVA challenge resulted in an obvious increase on the immunostained areas of TGF- β in the peribronchial tissues. However, the expression level of

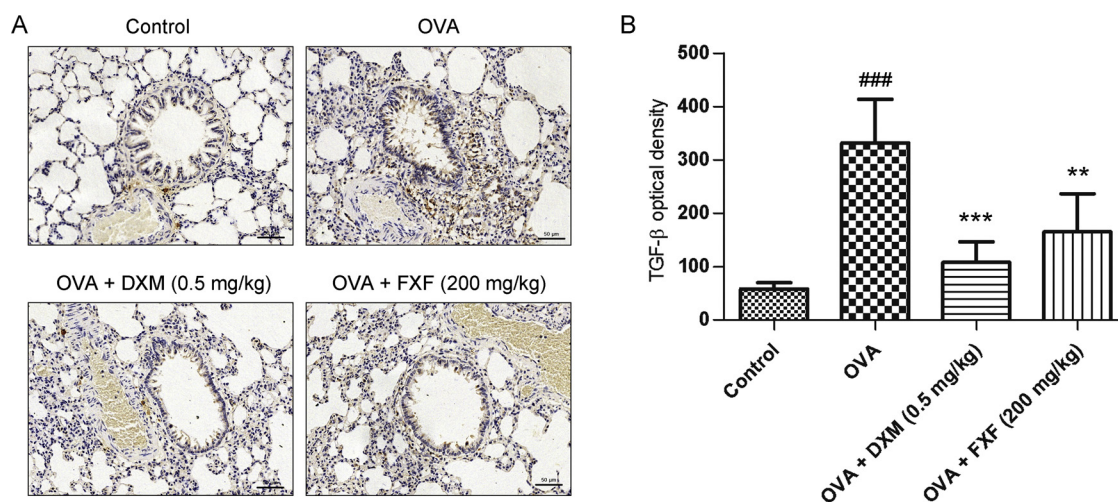


Fig. 8. The inhibitory effect of FXF on the expression of TGF- β and Smad-3 in lung tissues. (A) The amounts of active TGF- β were determined by immunohistochemical staining in the lung tissue from OVA challenged asthmatic rat. The positive staining of TGF- β was depicted in brown. (B) Semi-quantitative immunohistochemical staining analysis of TGF- β determining by an image analyzer. ### $P < 0.001$ denotes significant differences compared with the control group. ** $P < 0.01$ and *** $P < 0.001$ denote significant differences compared with the OVA-challenged group, respectively.

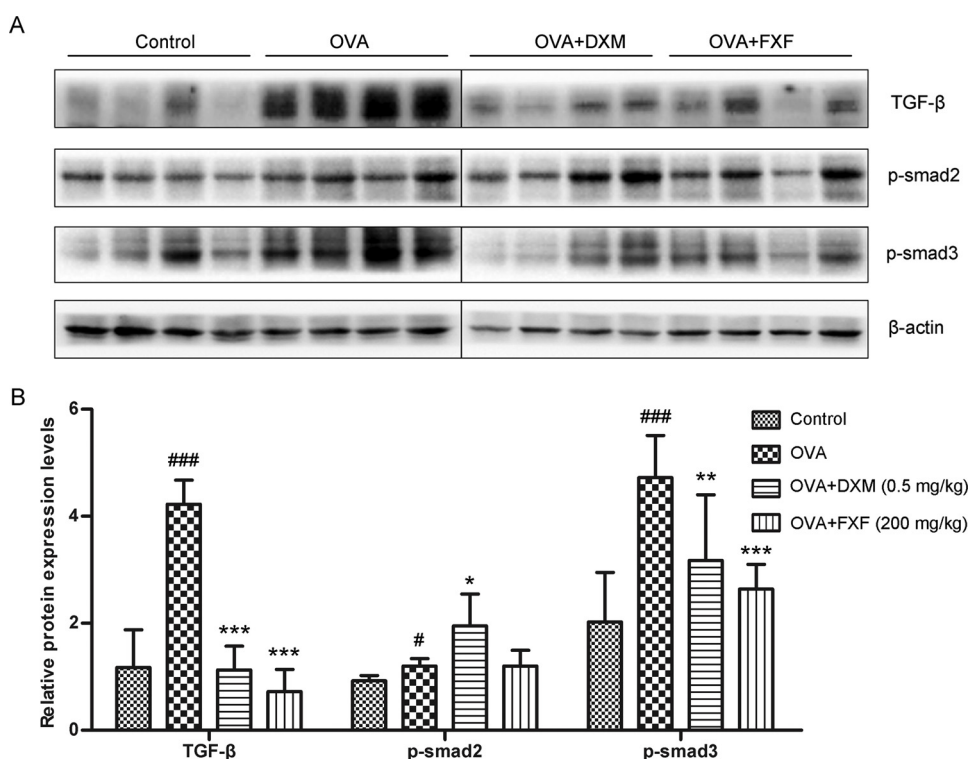


Fig. 9. Western blot analysis of TGF- β /Smad signaling pathway after the treatment of FXF in the OVA challenged rats. (A) Western blotting analysis of active TGF- β , p-Smad2 and Smad-3 protein expression levels in lung tissues. (B) Quantitative analysis of western blots results by gray intensity using the Gel-Pro Analyzer. The relative ratios of indicated protein expression were normalized to β -actin and were expressed as the mean values. # $P < 0.05$ and ### $P < 0.001$ denote significant differences compared with the control group. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$ denote significant differences compared with the OVA-challenged group, respectively.

TGF- β was markedly reversed by FXF treatment (Fig. 8A). The TGF- β optical density in OVA-challenged group was 332.2, while it reduced to 165.3 after the treatment with FXF (Fig. 8B). To determine whether TGF- β /Smad pathway inhibition was associated with the therapeutic effects of FXF, we further investigated the expression levels of TGF- β and its corresponding down-stream regulator p-Smad2 and p-Smad3 by western blotting (Fig. 9). Compared to the control group, the expressions of TGF- β , p-Smad2 and p-Smad3 in the lung tissues were markedly increased by OVA challenge, especially for the regulator TGF- β and p-Smad3. High dose of FXF treatment resulted in a significant decrease on the expressions of TGF- β and p-Smad3, while it exhibited little effect on the p-Smad2 expression (Fig. 9). According to the semi-quantitative analysis, the treatment of high dose of FXF on OVA-challenged rats decreased the expressions of TGF- β and p-Smad3 by 82.8% and 44.2%,

respectively (Fig. 9B). Overall, these data indicate that FXF could play protective roles on OVA-induced asthma rat model through the inhibition of TGF- β /Smad3 pathway.

4. Discussion

In this study, we investigated the protective role of the traditional herbal medicine Fangxiao Formula (FXF) on airway inflammation and remodeling in the asthma rat model. Our results demonstrated that FXF inhibited the airway inflammation by attenuating the inflammatory cells infiltration and decreasing the secretions of Th2 cytokines. In addition, FXF treatment resulted in reduced ASM thickness and PAS score, and decreased collagen deposition in the lung tissue, indicating improvement on the asthma airway remodeling. Furthermore, FXF

inhibited the protein expression levels of TGF- β and Smad-3 in lung sections, suggesting that FFX modulated airway inflammation and remodeling in asthma rat model via TGF- β /Smad signaling pathway.

Transforming growth factor- β (TGF- β) is a pro-fibrotic cytokine playing important roles in promoting the structural changes of airway wall, which closely associates with the development of airway remodeling and is highly expressed in asthma [20,21]. Smad proteins are important intracellular effectors of TGF- β regulating system, which could mediate transduction of intracellular signaling and promote transcription of target genes. The activation of TGF- β /Smad signaling pathway played central roles during the development of airway remodeling, which represented potential therapeutic target for asthma [22]. Previous studies have indicated that astragalus extract could improve the asthma airway remodeling via down-regulating the activity of TGF- β /Smad pathway, providing a potential drug target for asthma patients [11]. Resveratrol suppressed the TGF- β /Smad signaling pathway in the bronchial asthma mouse model, leading to the improvement on both inflammation and airway structural changes [1]. Our results demonstrated FFX was also able to block the TGF- β /Smad-3 pathway by decreasing the protein expressions levels of TGF- β and Smad3 phosphorylation.

Allergic asthma is closely associated with the increased pulmonary recruitment of inflammatory cells which was to a large extent driven by the activation of Th2 cells [23]. In the present study, FFX treatment significantly inhibited the secretions of Th2 cytokines in the BALF, reduced the number of eosinophils surrounding airways and improved the leukocyte infiltration in lung tissue of the asthma model. The inhibitory rates of 100 mg/kg FFX on the expression levels of IL-5 IL-9 and IL-13 ranged from 34.2% to 41.8%, which was consistent with its reduction effect on inflammatory cell recruitment in airway (Figs. 3 and 4). However, the treatment of FFX decreased the production of IL-4 by over than 50%, which appears to be slightly stronger than its effect on the inflammatory cell recruitment. Accumulating evidence has clearly demonstrated that not only Th2 cytokines but also other T cell related cytokines as well as epithelial cell cytokines and thymic stromal lymphopoietin (TSLP) are involved in the recruitment of inflammatory cells [24]. Therefore, the eosinophil trafficking and recruitment to inflamed airways could be regulated by multiple mediators beyond IL-4. Furthermore, the imbalance of Th1/Th2 lymphocyte cell contributed to the development of asthma pathogenesis [25]. The productions of Th2 related cytokines including IL-4, IL-5 and IL-13 were increased in asthma, which could induce eosinophil infiltration, excessive mucus and immunoglobulin E secretion [26]. In contrast to Th2 cells, The Th1 cell cytokines of IFN- γ and IL-12 could restrain the progress of asthma by antagonizing the Th2 cell responses and IgE synthesis [27]. Targeting the alternation of Th1/Th2 ratio is considered to be a promising approach to control the airway inflammation and has shown some exciting results for asthma [28]. Our results indicated that FFX inhibited the levels of Th2 cytokines (IL-4, IL-5, IL-9 and IL-13) and simultaneously enhanced the amounts of Th1 cytokines (IFN- γ), providing a novel approach to modulate the Th1/Th2 cytokine imbalance for asthma patients (Fig. 4). These observations suggested that the therapeutic effect of FFX toward asthma was closely related with the inhibition on airway inflammation and modulation of Th1/Th2 cytokine equilibrium.

Airway remodeling was the airway structural change in response to the chronic inflammation, which was an important pathophysiological feature of asthma. Increased airway smooth muscle (ASM) mass is a hallmark of airway remodeling which causes airway narrowing in asthma [29]. Our results demonstrated that the OVA challenged rats exhibited similar pathological changes of the bronchi and lung tissues with that observed in asthmatic patients, such as enhanced ASM thickness. FFX treatment significantly decreased the collagen deposition and inhibited the smooth muscle layer in the rat model, indicating a possibility to block the pulmonary fibrosis in asthma. Moreover, FFX containing serum obviously inhibited the TGF- β -induced proliferation

of human bronchial smooth muscle cells (BSMC) in vitro (Supplementary Fig. 2). In addition to BSMC, airway epithelial cell is also important for tissue remodeling which significantly contributes to the orchestration of pulmonary inflammatory and immune responses [30]. It would be very interesting to differentiate the effects of FFX on the proliferation of epithelial and smooth muscle cells present in the airways. Furthermore, it was observed that FFX dramatically inhibited the expression levels of α -SMA and PCNA in the OVA challenged rat model. Compared with DEX, high dose of FFX shows a similar ability to prevent the asthma airway remodeling. These results indicated the potential therapeutic effect of herbal medicine FFX on airway remodeling in allergic airways disease.

Traditional herbal medicines are mixture of several herbs with synergistic effects and reduced adverse effects, which have been widely used as safe and efficient approach for asthma treatment [31]. The decoction Ding Chuan Tang significantly improved the airway hyper-responsiveness and symptom in children with persistent asthma [32]. The Mahuang decoction could also mitigate airway inflammation via reduction on inflammatory regulator secretion and inflammatory cell infiltration through the regulation of IL-21 and STAT3 signaling pathway in rat asthma model [33]. Our results demonstrated the therapeutic effects of Fangxiao Formula (FFX) toward asthma. FFX is the crude aqueous extract from the herb mixture of eleven traditional medicines, which has the functions of supplementing qi, nourishing yin, tonifying lung and spleen, and calming the liver according to the traditional Chinese medicine theories. Six putative ingredients were identified from FFX decoction by comparing the retention time with those of standard compounds (Fig. 1). Most of these identified compounds from FFX decoction exhibited anti-inflammatory and immunomodulatory effects in the OVA-induced asthmatic model, which was at least partially responsible for the positive therapeutic effects. Naringin, a major compound from *Pericarpium Citri Reticulatae*, significantly reduced the enhanced cough, improved the airway hyper-responsiveness and inhibited the number of leukocytes in a guinea pig model [34]. Paeoniflorin, which is the principal active component from *Radices Paeoniae Alba*, exerts anti-asthmatic effects via modulation of the Th1/Th2 equilibrium [35]. However, due to the complexity and diversity of chemical compositions in traditional Chinese medicine, as well as the herb-herb chemical interactions during the preparation of FFX decoction, the main biological active compounds responsible for the therapeutic efficacy of FFX still need further investigation.

5. Conclusions

The current study reports that FFX obviously inhibited the airway inflammation by decreasing the inflammatory cell recruitment and regulating the imbalanced ratio of Th1/Th2 cytokines, and attenuated airway remodeling through regulation on ASM thickness, α -SMA and PCNA expression and collagen deposition in OVA sensitized asthma rat model. The remarkable therapeutic effect of FFX on asthma was closely related to the activity of TGF- β /Smad-3 signaling pathway. These observations suggested that Fangxiao Formula was effectively protective against allergic asthma and indicated the possibility of further developing FFX as potential therapeutic agent for asthma involving airway inflammation and remodeling.

Author contributions

Yuqing Ge conceived and designed the experiment, analyzed the data, interpreted the results and developed the manuscript; Rubin Cheng and Saijun Zhang prepared the powder of Fangxiao Formula and carried out the HPLC analysis; Siya Sun, Lan Li, Jianping Jiang and Chenxi Yang collaborated in the in vivo experiments, analyzed the data and discussed the findings; Rubin Cheng, Xiaobo Xuan and Jian Chen supervised the work and proofread the manuscript. All authors read and approved the manuscript.

Declaration of Competing Interest

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.biopha.2019.109429>.

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